What Context Compression Cannot Buy: Diagnosing the Recall Ceiling with a Frozen Backbone

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Abstract

Compression of a transformer's KV cache is usually reported as one number: the fraction of full-context quality retained. We ask what the missing fraction is made of. Training only 3.67M memory-slot parameters with the backbone frozen, we find that fraction is set by a single property of the task—how much of the answer can be copied from the context rather than inferred.

Splitting predicted tokens on that property separates them sharply: tokens that never appeared recover at +202%, above the verbatim cache they replace, while tokens seen exactly once reach +45%. Posed as a task—recover a random five-digit code hidden in the context—64 learned slots answer none of 150 trials and leave the model worse off than with no context at all, while 84 verbatim entries answer 98% when they are the right ones.

The same property says where each kind of memory pays. On held-out text, 64 learned slots are worth as much as 576 entries selected by H2O, and giving 64 of H2O's 576 entries to the slots recovers 101% of the gap, more than the full cache. On the needle task the slots are dead weight and everything rests on selection: SnapKV with the question in its window answers 99%, the same policies committing before the question arrives answer at most 20%, and our own retrieval, one set of positions shared by every layer, answers 12%. Across 0.6B to 14B each resource alone is flat, while what the slots add behind a fixed selection peaks at 1.7B–4B and halves by 14B.

We also quantify two measurement hazards: fine-tuning the backbone improves the no-context baseline by 0.58 nats, inflating every relative compression metric, and pooling scale points measured under different quantisation and corpora manufactured a clean monotone trend that a controlled sweep does not show. This version corrects a bug in version 1's attention scoring and every conclusion that rested on it.

1 Introduction

Work on shrinking the KV cache of a long-context transformer—by eviction, by low-rank or low-bit approximation, by learned summarisation into a fixed budget of slots—reports its result as a retention figure: this method keeps x% of full-context quality at $y\times$ compression. The figure is useful for choosing a method. It says nothing about the $1 - x$.

This paper is about the $1 - x$. We do not propose a compression method; several of the components we use were introduced elsewhere and we cite them as such (§7). We use a deliberately minimal setup—a frozen backbone with a few million trainable memory-slot parameters—as an instrument, and ask two questions about where compressed context loses to verbatim context.

Contribution 1: the ceiling is made of precise recall (§3). We group the tokens being predicted by whether that token appeared anywhere in the preceding context, i.e. by whether the

answer was available to copy (Table 1). The groups behave qualitatively differently rather than by degree. On tokens that never appeared, a 64-slot learned state recovers +202% of the gap between no context and full context—it is more useful than the verbatim cache it replaces, because it was optimised to predict the continuation whereas raw KV was not. On tokens that appeared exactly once, the same state recovers +45%. Since the copyable group is 79.6% of the suffix and carries 76.0% of the gap, the ceiling on abstractive compression is a property of the task, not a deficiency of the compressor. We then state the same claim as a retrieval task (§3.1), where the answer is a random code and abstraction is not merely insufficient but actively harmful. The same axis predicts where each kind of memory pays (§4.1): on natural text the learned state adds a fixed-budget gain behind any selector, including the strongest training-free one we ran, while on the needle task it is dead weight and the outcome rests on whether the selector sees the question and may choose per attention head. To our knowledge this decomposition has not been reported for compressed context; the nearest work stratifies tokens by n-gram novelty in *uncompressed* long-context models [Hu et al., 2024], or contrasts topical against unexpected needles at the task level [Deng et al., 2024].

Contribution 2: retrieval and compression are separable and roughly equal, and only the state's marginal value responds to scale (§4, §5). Pairing the abstract state with a small set of exactly-retrieved verbatim KV entries is a natural architecture and has been proposed before [Yuan et al., 2026, Jie et al., 2025]. What has not been measured, to our knowledge, is the attribution: we fix the retrieved entries and swap only the state behind them (absent / trained / randomly initialised), and independently fix the state and swap only the anchor *source* (a deployable query against an answer-aware oracle). This closes a 2×2 design in which each factor's contribution can be read holding the other fixed, and the result survives replacing our retrieval with a stronger training-free selector (§4.1). Agent-memory systems have compared "read the summary" against "fetch the source" [Chen et al., 2026, Ji et al., 2026], but with both the retrieval set and the summariser varying together. The scale axis separates them a second time: across five backbone sizes each resource is flat on its own, while what the learned state adds behind a fixed selection peaks at 1.7B–4B and roughly halves by 14B, behind three different selectors (§5). A single-resource account does not predict that.

Contribution 3 (a caveat, not a claim): two measurement hazards (§6). Relative compression metrics normalise by the distance between a no-context lower bound and a full-context upper bound. If the backbone is fine-tuned during compression training, the lower bound moves: we measure 0.58 nats of improvement on "predict the continuation with no context at all," shrinking the denominator by a third and inflating recovery from +8% to +66% on the identical configuration. This is not a new observation so much as a quantification of one: Kim et al. [2023] report the same phenomenon as a training pathology (the model learns to answer without consulting its memory) and freeze the LM in response, and Feldman and Artzi [2025] adopt an almost identical normalisation to control for questions answerable without context. Our contribution is to frame it as an *evaluation* confound with a measured magnitude, and to note the boundary: it affects relative metrics only, not absolute task accuracy. Second, we report seed variance, which is rarely reported in this literature and is largest exactly where it matters most. We also give two worked examples of what skipping this costs, both from our own earlier drafts of the scale comparison: mismatched prediction lengths put the gap at 0.492 against 0.822 nats and made the recovery figures incomparable, and pooling points measured under different quantisation and corpora produced a clean monotone trend that the controlled sweep does not show (§5).

2 Setup

Compressor. A context window of n tokens is consumed in segments. Each segment is encoded with the frozen backbone and reduced to k learned memory slots, which are prepended to the next segment's attention as synthetic KV entries. The only trainable parameters are the slot embeddings (0.03M) and, in the *deep* variant, a per-layer additive bias applied at memory positions only (3.67M, in the manner of prefix tuning [Liu et al., 2021]). Because attention is causal and the bias is restricted to memory positions, the backbone's behaviour on real tokens is bit-identical to the unmodified model.

Metric. For a held-out suffix we measure mean token cross-entropy under three conditions: UPPER (the full verbatim KV cache of all n preceding tokens), LOWER (no context at all), and the condition under test. We report

$$\text{recovery} = \frac{\text{LOWER} - \text{test}}{\text{LOWER} - \text{UPPER}}, \tag{1}$$

in nats. 1.0 means indistinguishable from the full cache and 0 means the context was worthless. **Values above** 1.0 **are attainable and are not an error**: the learned slots are trained to predict the next segment, whereas the verbatim cache is simply what the model happened to write down, so a compressed state can be more useful than the text it summarises. We call LOWER − UPPER the *gap*, and note that it is a property of the corpus rather than of the method: it is 1.269 nats on real dialogue and 0.286 nats on Project Gutenberg prose, meaning long context is worth little on coherent narrative to begin with (§8).

Anchors. In the hybrid conditions, a budget of a verbatim KV entries ("anchors") is selected from the offloaded full cache and paged back in alongside the k slots. Selection policies are compared in §4. Two details are load-bearing and easy to get wrong. First, any scheme that retains only part of the cache must pin the first few positions as an *attention sink* [Xiao et al., 2023]; without it recovery collapses to −400%. Second, positions must accumulate across segments rather than restart, which is worth 4 points.

Scale and cost. Qwen3-0.6B [Yang et al., 2025] for every measurement, with the hybrid architecture checked against 1.7B (§5); windows of 4096 tokens in §3 and 8192 in §4, stated per table because the two are not comparable. Everything ran on a single RTX 4060 Laptop (8 GB); training one memory module takes roughly 12 minutes, and no cloud compute was used for any number in this paper.

3 The ceiling is made of precise recall

Two levers that should raise a compressor's ceiling do not (§5): more training and more slots. At a fixed backbone, then, the residual is not something the compressor is failing to try hard enough at. The candidate is precise recall—the identifier introduced 2000 tokens ago, the number, the proper noun. Verbatim KV can be copied from. A summary cannot reconstruct a string it did not store. Scale does move the level (§5), and we return at the end of this section to what it does and does not change.

We test this by partitioning the predicted suffix. For each target token we ask whether that token appears anywhere in the preceding context, and compute per-token cross-entropy within

each group under all three conditions. Only the abstract state is provided—no anchors—so the comparison isolates what abstraction alone can serve.

target token	share	LOWER	UPPER	gap (nats)	recovery
never appeared	20.4%	5.731	5.518	**0.213**	+202%
appeared	79.6%	3.542	2.401	**1.141**	+58%
exactly once	9.6%	5.190	4.014	1.176	+45%
2–5 times	19.4%	4.539	3.045	1.494	+48%
6+ times	50.6%	2.848	1.849	0.999	+67%
all	100%	3.989	3.038	0.951	+65%

Table 1: Cross-entropy by token copyability, abstract state only (64 slots, 8 hops over 4096 tokens), mean of three independently trained modules. LOWER is no context, UPPER the full verbatim cache. Seed spread is 3.0–4.3 points on the large groups but 9.1 on "exactly once" (n=1176) and 11.3 on "never appeared", where the small gap in the denominator amplifies it—see the caution below.

Read the gap column first. The cleanest statement here does not involve our metric at all. Having the entire verbatim context is worth 1.141 nats on tokens that appeared in it and 0.213 nats on tokens that did not. Long context earns most of its value on material that can be copied: the copyable group is 79.6% of the suffix and carries 76.0% of the total gap, a figure identical across all three seeds because it depends only on the corpus and not on the module. Whatever a compressor does, this is the shape of what it is compressing.

Recovery, with a caution about the denominator. On tokens that never appeared, the 64-slot state recovers +202%—substantially more useful than the verbatim cache it replaces, because it was trained to predict the continuation whereas the cache is merely what the model wrote down. On tokens that appeared exactly once it recovers +45%, and recovery rises monotonically with how often the token was seen (+45 → +48 → +67). The caution: those extreme groups also carry the largest seed spread, and not by accident. Recovery divides by the gap; the never-appeared gap is 0.213 nats; dividing by a small number magnifies everything, including noise. The ordering is robust—+202 against +58 is a separation of 144 points against spreads of 11 and 4—but the value 202 itself is not stable to three significant figures, and we would not report it without the gap column beside it. This is a general hazard when a relative compression metric is evaluated on subsets, and a reason to prefer absolute nats when slicing.

The ordering is a property of the task; the levels are not. Repeating the split at 0.6B and 4B separates what is fixed from what is not.

Which tokens are hard does not change—the ordering holds at both scales, and tokens seen once remain the worst-served group by a wide margin. How hard they are does change, and in the direction we did not predict: the hardest column improves most (+25.7) and the easiest improves least (+10.9). So the ceiling is not a fixed property of abstractive compression. It is a property of what a *given* backbone can compress, and a larger one recovers a substantial part of what the smaller one loses on precisely the tokens we called incompressible. We therefore state the finding as the ordering rather than the level, and note that an extrapolation to "a large enough model would not need exact recall at all" is not supported by two points.

	once	2–5	6+	never	all
0.6B	+30.9	+40.1	+60.1	+126.7	+53.6
4B	+56.6	+56.7	+71.0	+165.2	+72.7
change	+**25.7**	+16.6	+**10.9**	+38.5	+19.1

Table 2: The same partition at two scales, single seed each. Two mismatches make the absolute values here lower than Table 1 and not comparable to it: these modules were trained at the shorter prediction length and evaluated over a full segment, and the 4B backbone is NF4-quantised while 0.6B is bf16. Only the pattern across columns should be read. 1.7B is omitted because that particular checkpoint diverged under recursion (§8).

What follows. Two readings. First, abstraction is not a degraded copy of verbatim context but a different thing—better on some tokens, much worse on others. Aggregate recovery figures approaching 100% should therefore not be read as "nearly as good as having the text," since they average over groups that behave in opposite directions. Second, the practical one: at a fixed backbone the residual lives in copyable tokens, and the way to buy it back is not a larger abstraction budget—§5 shows doubling it buys 0.6 points—but a small number of *exact* entries. §4 measures what that buys.

3.1 The same claim, stated as a task

Everything above is perplexity on held-out text, which is the form of evidence a reader has least reason to trust. The claim it supports—that what compression cannot buy is precise recall—can be stated as a task instead. We hide a uniformly random five-digit code in the context ("the secret code for warehouse 7 is 48213"), place the question at the end, and ask for the digits back. Copyability here is total rather than statistical: the code is drawn at generation time, so no prior over language helps, and a memory that did not store the digits cannot produce them.

memory available at question time	entries	recovery	exact match
full verbatim cache	4096	+100%	**100**%
no context	0	0%	0%
abstract slots	64	−**28.6**%	**0**%
anchors retrieved by the question	512	+14.0%	13%
the same + 64 slots	576	+13.2%	12%
SnapKV, question in its window	576	+99.6%	**99**%
oracle: the needle itself + sink + 64 slots	84	+94.4%	98%

Table 3: Needle retrieval, 4096-token contexts, 3 seeds × 5 needle depths × 10 samples = 150 trials per condition. Recovery is over the answer tokens only. The gap between no context and full context is 2.52 nats here against 0.95 on dialogue, which is what a task built entirely out of uncopyable answers looks like. "Anchors retrieved by the question" is our retrieval: last-layer attention from the question over the cached keys, one set of positions shared by every layer and head. SnapKV selects per layer and per KV head (§4.1).

Three readings.

The abstract state does not merely fail; it is worse than nothing. Sixty-four learned slots that recover most of the loss on ordinary text answer zero of 150 needles, and leave the model *worse*

off than having no context at all (−28.6%, consistent across seeds). The slots occupy attention while carrying nothing relevant to the question. This is §3's "exactly once" column taken to its limit, and it is the cleanest statement we can make of what abstraction cannot do.

Adding the state to retrieval buys nothing here. On dialogue the same state is worth +45 points behind the same anchors (§4); here it moves exact match by one point, inside the seed spread. Behind a selector that finds the needle the sign is clear: giving 64 of SnapKV's 576 entries to the slots lowers exact match from 99% to 97% and recovery by 4 points, where on text the same substitution gains 36 (§4.1). The value of a compressed memory is not a property of the compressor alone: it is set by how much of the answer the task allows to be inferred rather than copied.

Selection is the whole task, and it is solvable. Eighty-four entries answer 98% of needles when they are the right ones—the needle's own sixteen tokens and the attention sink, with the slots alongside. Our own retrieval, given the question as its query and 512 entries to spend, answers 13%, and only when the needle sits at depth 0.9, near the end of the context. SnapKV, reading the same question but selecting separately in every layer and every KV head, answers 99% at every depth. The failure is not in the signal—both policies see the question—but in how coarsely our policy turns it into a selection; we return to this in §4.1.

This is the counterpart to the diffuseness we report in §4, where a policy agreeing with the oracle on 10.5% of positions still captured four fifths of the value. Both are true, of different tasks: value is spread thinly over natural text and concentrated in a single span here, and a retrieval policy that is adequate under the first regime can be nearly useless under the second. We would expect any evaluation of KV-cache retrieval to move sharply depending on which regime its benchmark samples, and §3's decomposition is a way to say in advance which one a corpus is.

4 Separating retrieval from compression

The architecture under test pairs $k = 64$ abstract slots with $a = 512$ verbatim KV entries paged back from an offloaded full cache: a 576-entry budget against 8192 tokens, 14:1. Both halves are prior art (§7). The question here is only how the credit divides.

We vary two factors independently. *Anchor source* is either the deployable policy—a short speculative draft of the continuation, generated from the state and the attention sink alone, used as the retrieval query [Jie et al., 2025]—or an *oracle* that is allowed to see the tokens being predicted and selects the entries that actually help. *State* is either absent, or the rolling state that has compressed all 8192 preceding tokens, or a state that compresses only the most recent segment (§5 explains why the latter is better). All six cells use the identical retrieved entries within a row and the identical state within a column, so each difference is attributable.

The two contributions are roughly equal and close to additive. Reading along the first row of Table 4, adding the abstract state behind a fixed set of retrieved entries is worth +45.2 points; along the second row, behind the oracle's entries, +39.9. Reading down the last column, replacing the deployable retrieval with perfect selection is worth +20.2 points; down the first column, +25.5. Each factor is therefore worth slightly less when the other is already strong—the two memories are mildly redundant—but the interaction is small relative to either main effect: starting from retrieval alone at +32.6 and adding both main effects predicts +103.3 against an observed +98.0, an interaction of −5.3 points. Compression and retrieval are close to separate resources, and neither

anchor source	no state	rolling state	last-segment state
speculative (deployable)	+32.6	+74.7	+77.8
oracle (answer-aware)	+58.1	+96.1	**+98.0**

Table 4: Recovery at a 576-entry budget over 8192 tokens (14:1), mean of three independently trained memory modules. Seed spread is 1.6–4.8 points, the largest in the deployable no-state cell, whose anchors are retrieved by a draft generated from the (seed-dependent) state; the no-state oracle cell is identical across seeds to four decimal places, as it must be, since neither the oracle's selection nor the empty state depends on the trained module—a consistency check on the measurement rather than a result.

dominates. Both main effects are several times the largest seed spread in the table (4.8 points), which is what licenses reading them at all.

The budget is not the binding constraint; selection is. The best cell reaches +98.0% using 576 of 8192 entries—7% of the cache—so essentially everything the full cache provides is reachable at this budget *if the right entries are chosen*. The deployable policy reaches +77.8%. The remaining 20.2 points are a selection problem, not a capacity or a compression problem, which is worth knowing before spending effort on better compressors.

A detail sharpens this. The deployable policy's choices overlap the oracle's on only 10.5% of positions, yet it captures about four fifths of the available value. The value is therefore diffuse: many different subsets of the cache are nearly as good, and there is no small set of critical entries that a retrieval policy must find. That is encouraging for cheap heuristics and discouraging for the idea that a sharper query is the main thing missing.

Randomly initialised slots are worse than no slots at all. Substituting untrained slots for the trained state, holding the retrieved entries fixed, does not merely remove the state's contribution—it lands far below the no-state condition, between −13% and −45% across three runs against +32.6% with no state at all. (We report a range rather than a mean: the slots are re-drawn per window and the draw is not seeded, so this condition is the one place where run-to-run variation is expected and large. Its sign and magnitude are stable; its exact value is not.) Synthetic KV entries that lie outside the distribution the backbone expects actively corrupt attention over the real entries beside them. Two consequences: the +45.2 points above are a property of the trained state and not of merely occupying positions; and any scheme that writes fabricated entries into a frozen model's cache is working inside a narrow tolerance. Relatedly, every partial-cache condition here must pin the first few positions as an attention sink, without which recovery falls to roughly −400% [Xiao et al., 2023].

4.1 Against training-free selection at the same budget

The decomposition above says the learned state is worth +45 points behind fixed anchors. A reader is entitled to a blunter question: at the same memory, does any of this beat simply keeping the right 576 entries? Three well-known policies answer that without training anything, and the same policies let us ask the question the other way round—does the learned state still add anything behind a selector nobody would call a strawman?

at 576 entries, 4096-token context	recovery	seed spread
StreamingLLM [Xiao et al., 2023]: sink + recent window	+55.8%	—
SnapKV [Li et al., 2024b], per layer and head	+54.2%	—
H2O [Zhang et al., 2023], per layer and head	+77.9%	—
64 learned slots alone (64 entries)	+75.4%	1.2
ours: 512 speculative anchors + 64 slots	+88.6%	1.0
SnapKV, 512 entries + 64 slots	+90.2%	0.5
H2O, 512 entries + 64 slots	**+101.1%**	0.9
oracle anchors + 64 slots	+103.2%	0.6

Table 5: Identical budget on held-out dialogue, 12 windows. The training-free policies are deterministic given the model, so their figures are identical across the three seeds; rows with slots vary with the trained module. Every attention score here is the model's own attention, recomputed by hand and checked against the eager implementation at every layer.

The learned state adds a fixed-budget gain behind any selector. Spending 64 of the 576 entries on learned slots instead of on more selected entries adds +23.2 points behind H2O and +36.0 behind SnapKV, and H2O's selection with the slots behind it recovers *more* than the full cache does. Sixty-four slots on their own are worth about as much as 576 entries chosen by H2O (+75.4 against +77.9). This is §4's result restated against the strongest training-free selector we ran: retrieval and compression are separate resources of similar size, and pairing them beats spending the whole budget on either.

Our own retrieval is the weak component. Our full method recovers +88.6%, ahead of every policy that trains nothing but behind H2O's selection with our slots behind it. The difference is granularity. H2O and SnapKV choose separately for every layer and every KV head; our retrieval chooses one set of positions from last-layer attention and applies it to every layer, which is what paging verbatim entries back from an offloaded cache most naturally does. The same restriction costs H2O itself 29.6 points (+48.3% with a single shared set). A reader building on this architecture should retrieve per head.

What this does and does not establish. The comparison is not like-for-like and we would rather say so than let a reader discover it. The slots are trained—3.67M parameters for roughly 12 minutes on one 8 GB card—on the evaluation domain; the three policies train nothing and need no corpus, which is exactly what makes them deployable where our module is not (§8). We implemented all three from the mechanisms their papers describe rather than porting reference code, since the reference implementations are bound to particular serving stacks: SnapKV with a 32-token observation window and max-pooling of width 7, H2O with half the budget on the recent window and half on accumulated attention, both per layer and per KV head with scores summed over each head's query group.

On the needle task, the question decides. Table 6 runs the same policies on the task of §3.1. The eviction policies decide what to keep at prefill, before the question exists, and find almost nothing: H2O answers none of 150 needles, SnapKV 9%, and StreamingLLM 20%, all of it from the one depth at which the needle is recent. This is the query-agnostic failure reported for multi-turn use of compressed caches by Li et al. [2024a] and Kim et al. [2025], and we reproduce rather than

discover it. Given the question in its observation window—the way SnapKV is normally run, with the question closing the prompt—SnapKV answers 99%.

policy at 576 entries	sees the question	natural text	needle, exact match
StreamingLLM	no	+55.8%	20%
H2O, per head	no	+77.9%	0%
SnapKV, per head	no	+54.2%	9%
SnapKV, per head, question in window	yes	—	**99**%
the same, 512 + 64 slots	yes	—	97%
ours: 512 anchors + 64 slots	yes	+**88.6**%	12%
oracle anchors + 64 slots	—	+103.2%	98%

Table 6: The same policies on the two tasks. Needle figures are 3 seeds × 50 trials; per-seed exact match varies by at most 6 points. Natural text has no question to put in a window, so that row is not defined there. Our method sees the question because it keeps the full cache offloaded and retrieves at decode time; the eviction policies commit at prefill.

Our method sees the question too—it keeps the full cache offloaded and retrieves at decode time, which is the design's whole point—and answers 12%. The difference is again granularity, and here it is decisive. Averaged over layers and heads, SnapKV keeps only 39–50% of the code's digits at depths 0.1 to 0.7, and still answers every needle: the heads that read the answer keep it, and the rest spend their budget elsewhere. Our policy chooses one set of positions for all layers from last-layer attention, and at those depths it keeps at most 4% of the digits.

So the axis of §3 predicts two things about a compressed memory. Where the continuation is inferable, the learned state earns a large fixed-budget gain behind any selector. Where the answer is arbitrary, the state is dead weight and everything rests on selection—on whether the selector sees the request, and on whether it may choose per head. A method should be reported against the regime it targets; v1 of this paper reported the second regime with a broken selector, and the note on changes from version 1 at the end of the paper says what that changed.

5 What moves the ceiling, and what does not

lever	range	deployable	retrieval only
training on the recursion	init → 600 steps	+59.9 → +58.6	—
abstract slot budget	64 → 128 slots	+67.0 → +67.6	—
anchor budget	512 → 1024 entries	+89.2 → +96.8	—
backbone scale	0.6B → 14B	non-monotone, peak +**91.6**	see Table 8

Table 7: Training and abstraction budget do not move the ceiling; anchor budget does, by spending memory on verbatim entries. Scale moves it non-monotonically, rising to a peak at 1.7B–4B and returning by 14B to about its starting value, while each resource on its own stays flat across a 23-fold parameter range; Table 8 gives that sweep in full. The first two rows are earlier single-run experiments at their own settings; the anchor-budget row was remeasured for this version at 4096 tokens with three seeds (spread under one point). None of the first three rows is comparable to the fourth.

Training the recursion trades near for far. Six hundred steps of training dedicated to the recursion itself leave the 8-hop figure where it started (+59.9 → +58.6) while the 1-hop figure falls from +128.8 to +98.8. The module does not become better at carrying information across many hops; it becomes worse at the single hop in exchange for nothing. We read this as the loss occurring at each re-encoding rather than at the point of storage. Doubling the slot budget likewise buys 0.6 points, so neither capacity nor training is the binding constraint.

Scale moves the state's marginal value, not the components. Sweeping 0.6B to 14B under matched conditions separates quantities that a pooled comparison would confound. Every column below was trained in bf16 on the same corpus and evaluated, for this version, on one card.

	0.6B	1.7B	4B	8B	14B	8B (lr 2e-3)
each resource alone, 576 entries						
learned slots, all abstraction	+59.8	+53.2	+59.6	+59.3	+54.0	+57.7
verbatim entries chosen by rarity	+20.9	+26.9	+26.1	+26.9	+30.2	+26.9
verbatim entries chosen by H2O	+77.8	+80.2	+80.7	+80.6	+81.5	—
the learned state behind a fixed selection						
512 speculative anchors, no state	+46.3	+51.1	+55.9	+60.0	+62.8	+54.6
+ 64 slots (our method)	+81.5	+89.9	**+91.6**	+88.5	+82.6	+84.9
what the state adds	+35.2	**+38.8**	+35.7	+28.5	+19.8	+30.3
the same, in nats	0.344	**0.373**	0.349	0.276	0.203	0.293
512 entries by H2O + 64 slots	+96.3	+103.0	**+104.1**	+98.0	+94.0	—
what the slots add over H2O at 576	+18.5	+22.8	**+23.4**	+17.4	+12.5	—
gap (denominator, nats)	0.978	0.962	0.979	0.968	1.026	0.968
memory parameters	3.7M	7.3M	11.8M	18.9M	26.2M	18.9M

Table 8: Identical budget (576 entries) at five scales. Each column's slots come from one training run; the last repeats 8B at a lower learning rate. The upper block spends the budget on one resource and is flat. The lower block holds a selection fixed and adds the learned state behind it; what the state adds peaks at 1.7B–4B and roughly halves by 14B, behind both selectors and in absolute nats as well as recovery. SnapKV's selection gives the same shape (+29.5, +33.6, +32.1, +24.4, +19.6).

Figure 1 shows the shape. Each resource on its own is flat across a 23-fold range of parameters: 576 learned slots recover +53 to +60 everywhere, 512 rarity-chosen verbatim entries +21 to +30, and 576 entries chosen per head by H2O +78 to +82. What moves is what the learned state adds behind a fixed selection. Behind our speculative anchors it rises from +35.2 at 0.6B to +38.8 at 1.7B and falls to +19.8 at 14B; behind H2O's selection it rises from +18.5 to +23.4 at 4B and falls to +12.5. The same shape appears behind SnapKV and in absolute nats, so it is an artefact neither of one selector nor of the normalisation; the nats row is given because 14B's gap is 0.05 nats wider than the others'.

Our full method is the sum of this falling term and a rising one: the speculative anchors on their own improve steadily with scale (+46.3 to +62.8), since they are drafted from the state and read by a larger model. Its recovery therefore peaks in the middle (+91.6 at 4B) and returns at 14B to about where it started (+82.6 against +81.5 at 0.6B).

The two 8B runs differ in learning rate and seed. The first used the learning rate tuned at 0.6B, whose training curve peaked around step 500 and then declined—the signature of too large a step—and the checkpoint we must use is the final one; the second, at 2.5× lower learning rate, trained cleanly. They differ by 3.6 points on the full method, because the anchors are drafted from

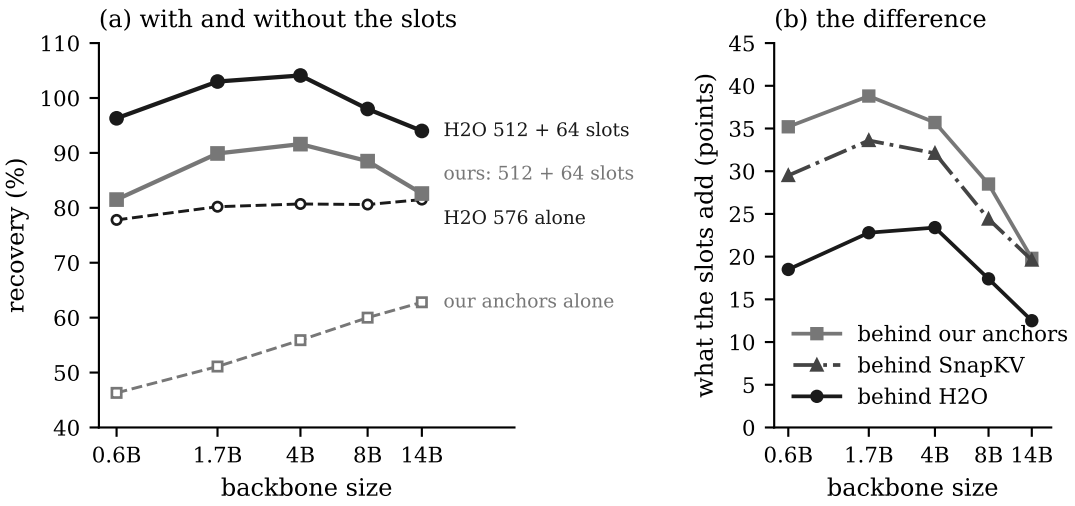


Figure 1: Table 8's lower block. (a) Solid lines add 64 learned slots to a fixed selection; dashed lines spend the same budget without them. (b) The difference between each pair, with SnapKV's selection added: what the slots add rises to 1.7B–4B and then falls, behind all three selectors.

the state and so differ between runs, but by 1.8 points on what the state adds behind them (+28.5 and +30.3), which is the quantity the argument rests on.

What scales, then, is neither memory on its own but the marginal value of the learned one. We do not have a mechanism, and the obvious candidate does not survive contact with the data. The memory module's parameter count grows with the backbone (3.7M to 26.2M) against a fixed training budget, so under-training would predict the *smallest* module to add the most—but 0.6B is not the peak, which sits at 7–12M parameters with both ends lower. Something about the middle of this range suits a 64-slot summary beside exact entries, and five points do not say what.

One further observation, offered as a lead rather than an explanation: randomly initialised slots are catastrophic at 0.6B (−64.7%, against +46.3% for no slots behind the same anchors) and nearly harmless at 14B (+60.9% against +62.8%). Larger backbones tolerate arbitrary synthetic KV entries that smaller ones cannot read. Whatever makes a large model forgiving of noise in its cache may also make it less responsive to the structure a trained compressor puts there.

Reading this result carefully. Two cautions, because the shape of this curve invites over-reading. First, each scale point is a single training run apart from 8B; at 0.6B the seed spread is 1.8 points on the full method and 4.5 on what the state adds (Table 4, at 8192 tokens), below the 19-point decline here but an extrapolation to the larger points. Second, and more important for anyone extending this: we obtained a clean monotone increase from an earlier, pooled version of this comparison, in which the 4B point came from a different machine, a different quantisation and a different corpus than the others. Each individual correction was small—1.2 points for NF4, 0.3 for the corpus—and together they manufactured a trend that the controlled sweep does not show. Small, individually defensible corrections compose into artefacts.

The effective window is about one segment. A rolling state that has passed over all 8192 tokens recovers +61.5%; compressing *only the most recent 512* recovers +65.6%. Carrying the state forward therefore dilutes rather than accumulates, and the distant past is better served by anchors than by a longer chain of compression. Recurrent compressive memories are known to concentrate

on the recent past [Rae et al., 2019, An et al., 2025]; the contribution here is that on this task the rolling variant is not merely no better but worse than not rolling at all, which changes what one should build.

6 Two measurement hazards

6.1 Fine-tuning the backbone deflates the denominator

	UPPER	LOWER	gap	recovery
backbone frozen	3.459	3.951	0.492	+**66**%
LoRA enabled	3.037	**3.370**	**0.332**	+8%

Table 9: Identical learning rate, identical step count; the only difference is whether the backbone may adapt. The adapter improves "predict the continuation from nothing" by 0.58 nats, removing a third of the denominator in Eq. 1.

The adapter is not learning to compress. It is doing language-model fine-tuning on the evaluation distribution, and one side effect is that the task gets easier. Note the direction of the trap: here it made the frozen number look better, because the same slots were measured against an undeflated denominator. In general the sign depends on which configuration is being promoted, which is precisely why the confound is worth naming.

Scope. This affects relative metrics of the form Eq. 1 only. Work reporting absolute downstream accuracy is unaffected, and work that fine-tunes both a compressor and a reader for deployment is doing something legitimate that this critique does not touch. See §7 for the closest prior statements.

6.2 Variance is largest where the headline is

configuration	seed 0	seed 1	seed 2	spread
16 slots (32:1)	+15	+41	+66	**51 points**
32 slots (16:1)	+105	+66	+103	**39 points**
72 slots	+114	+109	+111	5 points
128 slots	+115	+111	+112	4 points
hybrid, six conditions (§4)				1.6–4.8 points

Table 10: Identical data and hyperparameters; only the random seed differs. The first four rows are single-hop recovery measured during training, which is why they exceed 100% and are not comparable to the multi-hop figures elsewhere; what matters is the spread down each row. The last row covers the six cells of Table 4, excluding the randomly initialised state, whose slots are re-drawn per window and unseeded by construction.

The compression ratios that are most attractive to report are the ones whose results are least reproducible. We found this the hard way: a checkpoint was lost and retrained under identical settings, and the 16-slot figure moved from +83% to +38%. Any claim in this area resting on a few points—including several we had provisionally drawn about scale—requires its own spread before it

means anything. We recommend reporting seed spread at the highest compression ratio claimed, and we note that the hybrid architecture is not merely better on average but an order of magnitude more stable, which may be the more important property.

7 Related work

Our instrument is assembled from published components; what follows separates what we borrowed from what we are claiming.

Discarding or approximating real cache entries. One family keeps the cache made of real entries and decides which to drop. Zhang et al. [2023] retain a recent window plus persistent heavy hitters identified by accumulated attention; Li et al. [2024b] select using an observation window at the end of the prompt; Cai et al. [2024] allocate unequal budgets across depth. Xiao et al. [2023] show that the first few positions act as an attention sink whose removal is catastrophic—a result we depend on operationally, since every partial-cache condition in this paper would otherwise collapse. Liu et al. [2024] approximate rather than evict, quantising the cache to two bits. Tang et al. [2024] and Xiao et al. [2024] keep the full cache but attend sparsely, scoring blocks or representative tokens to decide what to load; Quest in particular selects per attention head from the current query, which is the granularity our own retrieval lacks (§4.1). These methods never fabricate entries, so the distribution constraint we report in §4—that untrained synthetic slots are worse than no slots—does not arise for them. We compare against three of them at a matched budget, alone and with our slots behind them, in §4.1.

Learning to compress context into slots. The second family replaces spans of context with a small number of learned vectors: gist tokens [Mu et al., 2023], autoencoding into memory slots [Ge et al., 2023], summary vectors trained by continued adaptation [Chevalier et al., 2023], and aggressive prompt compression [Li et al., 2024c]. Recurrent variants compress repeatedly as the sequence advances [Rae et al., 2019, Bulatov et al., 2022, An et al., 2025]. Our compressor is a minimal member of this family—learned slot embeddings plus a per-layer bias at memory positions in the manner of prefix tuning [Liu et al., 2021]—and we make no claim for it. We hold the backbone frozen throughout, which within this family is closest to Ge et al. [2023], Pilchen et al. [2025] and Li et al. [2024c].

Pairing compressed memory with exact entries. The hybrid we measure has been proposed. Yuan et al. [2026] combine compressed distant slots with exact recent entries for speculative decoding, discarding the distant verbatim cache rather than retrieving it. Jie et al. [2025] keep verbatim KV on the CPU and prefetch it by predicting what the next step will attend to, which is the speculative-retrieval mechanism we use as our deployable policy. At the agent-memory level, Chen et al. [2026] and Ji et al. [2026] both contrast reading a compressed representation against fetching the source, and both find neither alone suffices—consistent with our decomposition, though at the level of task accuracy with retrieval set and summariser varying together, rather than holding one fixed to attribute the difference. We treat all four as prior art for the architecture and confine our claims to the attribution.

Whether to freeze the reader. Freezing is usually motivated by plug-and-play reuse or by preserving abilities outside the compression task [Ge et al., 2023, Pilchen et al., 2025, Li et al., 2024c]; Rau et al. [2024] argue the opposite for retrieval-augmented generation, holding that a

frozen decoder cannot make good use of compressed context, and Szilvasy et al. [2026] continue pretraining the model jointly with their utility predictor. Closest to §6, Kim et al. [2023] report that naively fine-tuning the whole model during compression training lets it answer without consulting the memory at all—their no-context training loss falls from 2.69 to 1.84 while test loss stays at 2.59—and freeze the LM in response. They frame this as overfitting on inputs, i.e. a training pathology. Feldman and Artzi [2025] adopt a normalisation nearly identical to Eq. 1 in order to discount questions answerable without context, but fine-tune adapters on both sides and present the correction as a property of the question set. Our addition is narrow: the same effect read as an *evaluation* confound, with a magnitude attached (0.58 nats, a third of the denominator) and a stated scope. We do not claim the principle is new.

Compressibility depends on the task. That compression costs vary sharply across task types is established [Agrawal and Mayer, 2026, Liu et al., 2025]. Our dialogue-versus-prose gap figures (1.269 against 0.286 nats) are reported in that spirit—as a caution about which corpora can support this measurement at all—and not as a finding.

Per-token analyses of long context. Nearest to §3, Hu et al. [2024] stratify token-level perplexity by how often an n-gram occurs in the added context and find that newly appearing material correlates with the benefit from a longer window, but they study *uncompressed* long-context models, so the question of what compression specifically destroys does not arise. Deng et al. [2024] analyse gist-based compression at finer grain than task accuracy and find unexpected information is lost preferentially—topically relevant word needles at 50.7 against surprising ones at 35.8 under 8× compression—which points the same direction as our result while grouping by a needle's relation to the topic rather than by whether the target token was copyable. To our knowledge the grouping we use, over the tokens actually being predicted under compression, has not been reported; §8 states the limits of that claim.

8 Limitations

- **Mostly perplexity, and one synthetic task.** Apart from §3.1 every number here is token cross-entropy on held-out continuations. The needle task is synthetic and of our own construction; it establishes the endpoint of the copyability axis rather than performance on anything anyone wants done. Established suites—LongBench, RULER—would test whether the two regimes we describe appear at the mix of copyable and inferable answers that real tasks have, and we do not run them.
- **One model family, five scale points, one repeated.** Qwen3 at 0.6B, 1.7B, 4B, 8B and 14B in a single matched sweep, with 8B repeated at a second learning rate. The seed spread underpinning the comparison was measured at 0.6B and extrapolated upward, and every point except 8B is a single run. Larger dense points would show where the decline ends; MoE backbones would test whether it is specific to dense models, but that introduces an architecture difference this sweep deliberately avoids. All five points were trained in bf16 with the same corpus and settings (14B on a second, larger rented card) and re-evaluated for this version on a single card; an earlier, pooled version of this comparison mixed NF4 and a different corpus, and §5 describes what that did.
- **An instability we could not pin down.** In an earlier 1.7B configuration the rolling state diverged under multi-hop recursion—−101.5% and −47.3% on two seeds—while single-segment

states from the same checkpoints behaved normally. The 1.7B point in the controlled sweep (Table 8) shows no such divergence (+53.2% on the same condition), so whatever causes it is not the scale point itself; the runs differ in corpus, host and PyTorch version, and we did not isolate which. We mention it because anyone reproducing the rolling conditions may encounter it, and because it is the reason we report the sweep from single-segment states.

- **Our retrieval is the weak component.** It selects one set of positions for every layer from last-layer attention, and per-head selection beats it on both tasks: H2O's selection with our slots behind it recovers +101% against our +88.6% on text, and SnapKV answers 99% of needles against our 12% given the same question (§4.1). We did not rebuild the method around per-head retrieval; anyone using this architecture should.

- **The baseline comparison is not like-for-like.** §4.1 matches memory budget, not training cost: H2O, SnapKV and StreamingLLM require no training step and no corpus, while our module is trained on the evaluation domain and is domain-specific. The table says what a small learned component buys at a fixed budget; it does not say those policies are badly designed, and a deployment that cannot train on its own traffic is one where they win by default. Our H2O and SnapKV are reimplementations from their papers' descriptions, not the reference code, and reference implementations differ in details such as pooling and per-layer budgets.

- **Corpus.** The dialogue corpus is a private collection of real conversation logs and cannot be released. Gap figures for a public prose corpus are given for contrast, and the low gap there (0.286 nats) is itself a warning about which corpora can support this kind of measurement.

- **The literature search is not exhaustive.** Our "to our knowledge" claims rest on a bounded survey: a written brief stating the two questions and the finding we expected to be pre-empted was given to a search-capable model, whose returned claims were then re-checked, and every arXiv identifier cited here was subsequently resolved against the arXiv API to confirm that the paper exists and that its title and first author match. That procedure establishes what we read, not what exists; several of our conclusions did turn out to be pre-empted and are cited as such in §7. We phrase the remaining two as "to our knowledge" rather than as priority claims, and would rather be corrected on them than defend them.

Changes from version 1

Version 1 (10.5281/zenodo.22897854) contained a bug and several errors, corrected here.

The bug. Every policy that selects context by attention—our deployable retrieval and our implementations of H2O and SnapKV—recomputes attention by hand, since the fused kernel does not return weights. Version 1 did so without applying the rotary position embedding to the query, while the cached keys carry it. The scores were therefore not the model's attention: on a test sequence their top 32 positions agreed with the model's own on one. This version rotates the query and checks the hand-computed attention against the eager implementation at every layer before any experiment runs. Oracle selection, which matches tokens rather than scoring attention, and every result that involves no retrieval—the copyability decomposition, the abstract-state figures, the fine-tuning confound—were unaffected and are unchanged.

What changed as a result.

- **Baselines.** H2O and SnapKV now select per layer and per KV head, as specified; version 1 applied one set of positions from the last layer to every layer and ran SnapKV's observation window without its preceding context. Version 1's comparison (+89.5% against a best baseline of +55.8%) is replaced by Table 5: +88.6% against H2O's +77.9%, and H2O's selection with our slots behind it at +101.1%.
- **The needle comparison.** Version 1 reported that our method placed last on the needle task because its query was a drafted guess. That was wrong independently of the bug: on the needle task our retrieval already used the question, and our SnapKV never saw it. With correct scoring the outcome is decided by whether the selector sees the question and selects per head (Table 6); version 1's Figure 2 is withdrawn.
- **Decomposition.** The deployable row of Table 4 moves from +31.4, +71.9, +75.4 to +32.6, +74.7, +77.8; the oracle row is unchanged and so are the conclusions.
- **Anchor budget.** The checkpoint behind version 1's +87.8 → +96.6 row no longer exists; the row is remeasured with the three checkpoints used elsewhere.
- **Scale.** The sweep was re-evaluated on one card. The full method's row moves from +82.6, +90.2, +90.1, +83.3, +75.9 to +81.5, +89.9, +91.6, +88.5, +82.6: the peak stays at 1.7B–4B, but 14B no longer falls below 0.6B. Rows that involve no attention scoring reproduce version 1 to within 0.2 points on the new card. Version 1 also measured the hybrid's gain against anchors chosen by rarity rather than against its own anchors; Table 8 now reports what the state adds behind the same entries, which peaks at 1.7B–4B and roughly halves by 14B behind three selectors. Version 1's statement that the two 8B runs agree within a point holds for that quantity (1.8 points) but not for the full method (3.6).
- **Mislabels.** The oracle row of version 1's needle table read "512 anchors"; it used the needle's own 16 tokens and the 4-token sink, beside the 64 slots. The limitations text still described an earlier plan (4B under NF4, 8B missing) that the controlled sweep had already superseded.

Use of AI assistance

All experiments, analysis, debugging (including identifying the attention-scoring bug that necessitated version 2) and scientific conclusions in this paper are the author's own work. An AI writing assistant (Anthropic's Claude) was used to help draft and edit the English prose. The author reviewed and takes responsibility for the final text.

Reproducibility

Code, logs and the full experimental record—including the intermediate conclusions that were later falsified—are at `https://github.com/HoshiGT/frozen-kv-memory`. The dialogue corpus is withheld for privacy reasons; the prose corpus and its preparation script are included.

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